TECHNICAL TRAINING CATALOGUE 2026

OUR COMMITMENT:

TO ASSIST YOU IN DEVELOPING THE COMPETENCE AND PERFORMANCE OF YOUR EMPLOYEES AND YOUR SERVICE.

(Version of October 27, 2025)

EDITO

Dear Subsidiary,

The Group Academy is pleased to present its catalogue of services for 2026 and to inform you of our commitments for all subsidiaries:

- ✓ **To support you in developing the skills and performance of your employees and your service,**
- ✓ **To support you in resolving technical problems on customer vehicles,**
- ✓ **To standardize and normalize the skill levels of the group's technicians.**

The diversity of brands deployed by CFAO is a real asset for the development of our skills development offers to subsidiaries throughout the company's territories. This diversity challenges us to ensure a high level of service quality within the after-sales service, the parts service, the sales department and the financial and legal service.

To improve the quality of your employees' services, we recommend deploying the following technical skills acquisition cycle:

- ✓ **Measure** the current skills of technicians and teams,
- ✓ **Exploit** the results to guide the skills development strategy,
- ✓ **Define** individual and collective skills development plans,
- ✓ **Acquire** new skills through face-to-face or remote trainings,
- ✓ **Certify** the new skills acquired,
- ✓ **Apply** new skills in a professional environment,
- ✓ **Perform** in a professional environment.

We also offer to **assist** you in **resolving technical problems** that you encounter in your workshops, with **technical support integrated** into the Academy.

The development of new skills is carried out by certified experts each in their areas of expertise, using innovative training methods.

Thank you for your commitment to developing the skills of your employees. We look forward to hearing from you.

Laurent GONIN

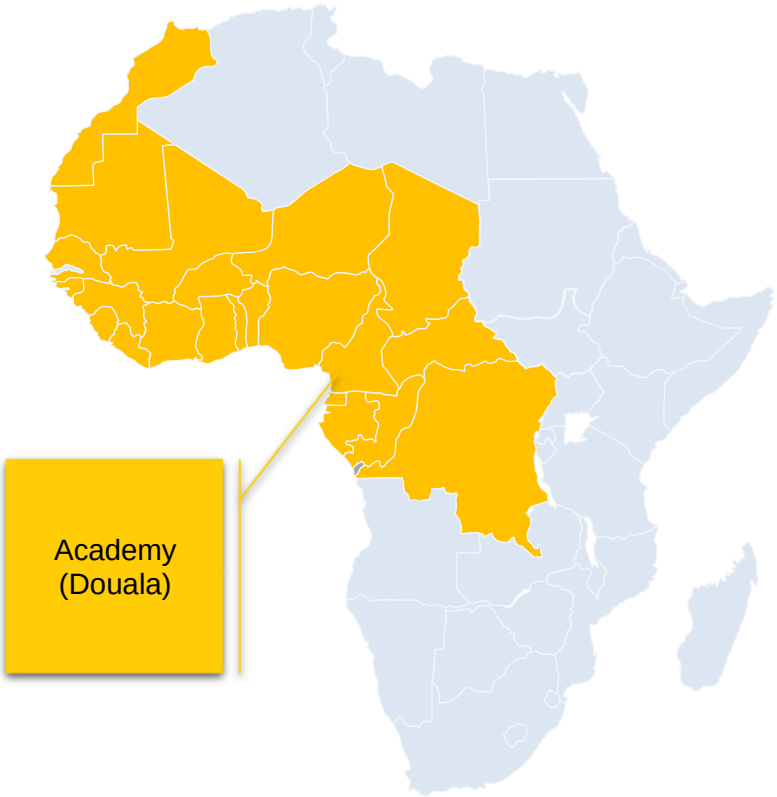
TEAM EQUIPMENT

The Douala Academy for EQUIPMNET is dedicated to the **RENAULT TRUCKS, JCB, TOYOTA FORKLIFT, BT, FUSO, HINO, SINOTRUK, KING LONG, SHACMAN and LOVOL** brands and is equipped with:

- A reception and a restaurant,
- Two training rooms (board, video projector, others),
- A 200 m² workshop,
- Didactic equipment for engines, transmissions and others,
- A RENAULT TRUCKS K Range truck

Our training courses are delivered at the Douala Academy and at all our subsidiaries. We also provide training at our customers' sites. Training sessions are made up of a maximum of twelve trainees.

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CFAO Training Director



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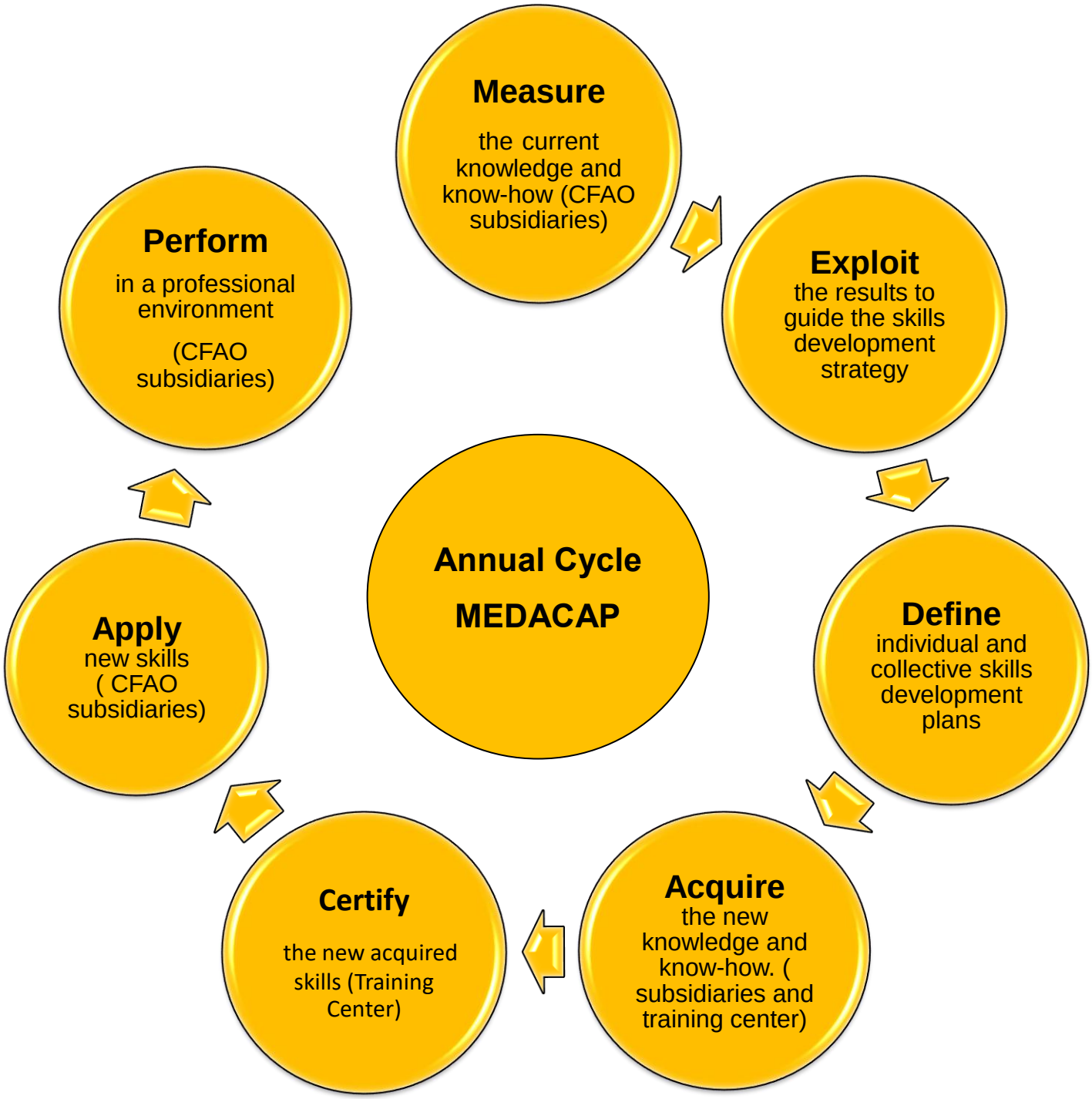


Thierry KOUGHA
Technical Trainer Expert (TOYOTA FORKLIFT, JCB, LOVOL)

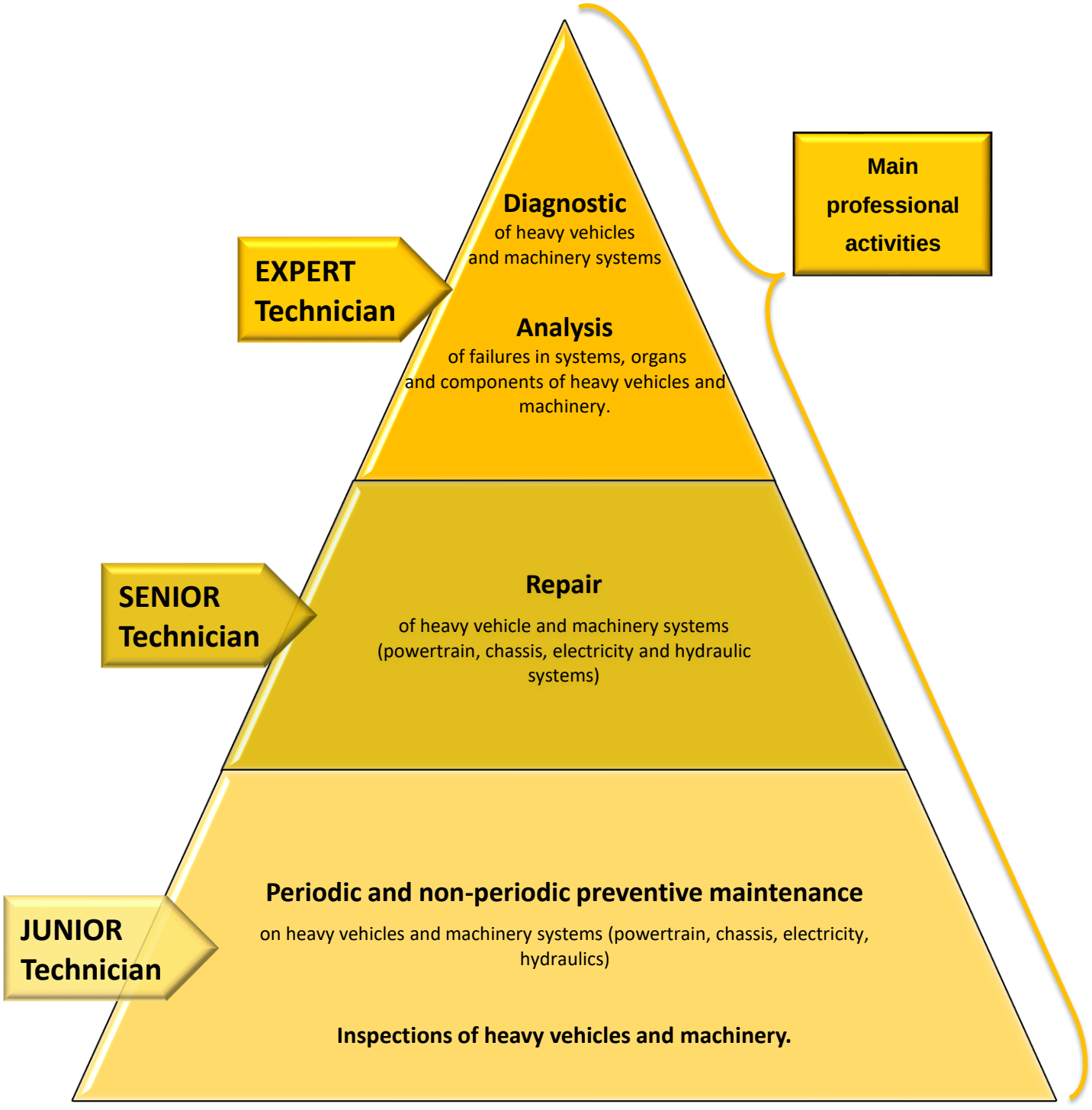


Roland NENDA
Operator Trainer
All- brands expert (Trucks and Heavy Equipment)

DEVELOPING TECHNICIANS' PROFESSIONAL EXPERTISE



SKILL LEVELS OF TECHNICIANS



TRAINING BY TECHNICIAN LEVEL

	Theoretical training (Knowledge - > E-Learning/Distance learning)		Practical training (Know-how -> Face-to-face)	
	Thematics	**Duration (days)**	**Thematics**	**Duration (days)**
Technicien EXPERT	❖ The 7 stages of diagnostics applied to different machine systems (powertrain, chassis, electricity), ❖ Features of the Diag tool	10	❖ Diagnostic of electrical/electronic components on JCB machines,	5
Technicien SENIOR	❖ Technological knowledge of the various machine systems (powertrain, chassis, electricity). ❖ Removal and refitting procedure in accordance with the manufacturer's recommendations, ❖ Disassembly and reassembly procedure in accordance with the manufacturer's recommendations, ❖ Procedure for checking components according to manufacturer's recommendations.	30	❖ Repair of POWERSHIFT PS750/764/766 gearbox, ❖ Repair of SYNCHROSIFT SS 600/700 gearbox, ❖ Repair of POWERSHIFT 3/4/6 WG ERGOPOWER gearbox, ❖ Repair of DIESELMAX 444 TIER 2/3 Engine with mechanical and electronic injection, ❖ Repair of Hydraulics for JCB machines.	5 5 5 5 5
Technicien JUNIOR	❖ Brand History (JCB ranges, characteristics, etc.), ❖ Description of the machine structure and roles of the different machine systems (powertrain, chassis, electricity). ❖ Role of fluids and lubricants. ❖ Principles of use of metrology tools, ❖ Periodic and non-periodic preventive maintenance on JCB machines, ❖ Fundamental principles of the engine, transmission and electricity, ❖ Others.	15	❖ Preventive maintenance of JCB backhoe loaders, ❖ Preventive maintenance of JCB wheel loaders, ❖ Preventive maintenance of JCB tracked excavators JS200 series, ❖ Preventive maintenance of JCB wheeled excavators JS100W Series, ❖ Preventive maintenance of telescopic handlers JCB 540-170 series. ❖ Preventive maintenance of JCB handlers RTFL940.	5 5 5 5 5 5

CONTENTS

Page	Code	Training designation	Date	Number of days	Location
8 to 12		JUNIOR TECHNICIAN (PREVENTIVE MAINTENANCE)			
9	JCB01	Preventive maintenance of JCB backhoe loaders,	To be defined	5 days	To be defined
10	JCB02	Preventive maintenance of JCB wheel loaders,	To be defined	5 days	To be defined
11	JCB03	Preventive maintenance of tracked excavators JCB JS200 series,	To be defined	5 days	To be defined
12	JCB04	Preventive maintenance of JCB JS100W series wheeled excavators,	To be defined,	5 days,	To be defined
13	JCB05	Preventive maintenance of telescopic handlers JCB 540-170 series.	To be defined	5 days	To be defined
14 to 19		SENIOR TECHNICIAN (REPAIR)			
15	JCB07	Repair of POWERSHIFT PS750/764/766 gearbox,	To be defined	5 days	To be defined
16	JCB08	Repair of SYNCHROSIFT SS 600/700 gearbox,	To be defined	5 days	To be defined
17	JCB09	Repair of POWERSHIFT 3/4/6WG ERGOPOWER gearbox,	To be defined	5 days	To be defined
18	JCB10	Repair of DIESELMAX 444 TIER 2/3 engine with mechanical and electronic injection.	To be defined	5 days	To be defined
19	JCB11	Repair of Hydraulics on JCB machines	To be defined	5 days	To be defined
20 to 21		EXPERT TECHNICIAN (DIAGNOSTIC)			
21	JCB12	Electrical/Electronic Diagnostic of JCB Machines	To be defined	5 days	To be defined

JUNIOR TECHNICIAN

PREVENTIVE MAINTENANCE



BACKHOE LOADER



WHEEL LOADER



TRACKED EXCAVATOR



WHEELED EXCAVATOR



ROUGH TERRAIN FORKLIFT



TELESCOPIC HANDLER

JCB01	PREVENTIVE MAINTENANCE OF JCB BACKHOE LOADER

Level, mode & duration	JUNIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, chassis, electricity, hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (powertrain, chassis, electricity, hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Backhoe loader ranges and their characteristics, • Description of machine systems (Powertrain, Chassis, electricity, hydraulics, air conditioning, Steering, Brakes,) and their components, • Principle of use of metrology tools, • The different types of maintenance applied to backhoe loaders and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations. ► **Practical part :** • Replace consumables (Engine, Transmissions SS 600/700, Hydraulics), • Replenish oils and fluids (Engine, Transmissions SS 600/7000, axles, hydraulic systems and others), • Check that machine components (dimensions, flatness, voltage, current, tightness, pressure, etc.) comply with the manufacturer's recommendations, • Remove and refit components (filters, belts and accessories, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the backhoe loader systems, • Reset the maintenance hour meter after operations in accordance with the manufacturer's recommendations. ► **Evaluation part :** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-Learning and/or classroom training in Mechanics, Electricity and Basic Hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, Adapted machine(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

JCB02	PREVENTIVE MAINTENANCE OF JCB WHEEL LOADER

Level, mode & duration	JUNIOR	Face-to-face	05 days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, chassis, electricity, hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (powertrain, chassis, electricity, hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Ranges of JCB wheel loaders and their characteristics, • Description of machine systems (Powertrain, Chassis, electricity, hydraulics, air conditioning, Steering, ...) and their components, • Principle of use of metrology tools, • The different types of maintenance applied to wheel loaders and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations. ► **Practical part :** • Remove and refit components (Filters, belt and accessories, Hydraulic hose, etc...) in accordance with the manufacturer's recommendations. • Replace consumables (Engine, BV ZF WG, axles, Hydraulics, etc.). • Replenish oils and fluids (Engine, BV ZF WG, axles, hydraulic systems and others) • Check that components comply with the manufacturer's recommendations (dimensions, flatness, voltage, current, tightness, pressure, etc.). • Visually inspect the loader's systems, • Reset the maintenance counter after work in accordance with the manufacturer's recommendations. ► **Evaluation part :** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Preventive Maintenance Technicians		
Prerequisites	E-learning and/or classroom training in Mechanics, Electricity and Basic Hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, Adapted machine(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

JCB03	PREVENTIVE MAINTENANCE OF JCB TRACKED EXCAVATORS JS200 SERIES

Level, mode & duration	JUNIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, chassis, electricity, hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (powertrain, chassis, electricity, hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Ranges of tracked excavators, JS200 series and their characteristics, • Description of machine systems (Powertrain, Chassis, electricity, hydraulics, air conditioning, Steering, ...) and their components, • The different types of maintenance applied to tracked excavators and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations. ► **Practical part :** • Replace consumables (Engine, ZF HL Transmissions, Hydraulics), • Replenish oils and fluids (Engine, ZF HL Transmissions, axles, hydraulic system and others), • Check that machine components (dimensions, flatness, voltage, current, tightness, pressure, etc.) comply with the manufacturer's recommendations, • Remove and refit components (filters, belts and accessories, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the systems on tracked excavators JS200 series, • Reset the maintenance counter after operations in accordance with the manufacturer's recommendations. ► **Evaluation part :** • Initial test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-Learning and/or classroom training in Mechanics, Electricity and Basic Hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, Adapted machine(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

JCB04	Preventive maintenance of JCB JS100W series wheeled excavators,

Level, mode & duration	JUNIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, chassis, electricity, hydraulics) in operation in accordance with the manufacturer's recommendations, • **Inspect** the machine's systems (powertrain, chassis, electricity, hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Ranges of tracked excavators, JS100W series and their characteristics, • Description of machine systems (Powertrain, Chassis, electricity, hydraulics, air conditioning, Steering, ...) and their components, • The different types of maintenance applied to tracked excavators and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in compliance with the manufacturer's recommendations. ► **Practical part :** • Replace consumables (Engine, ZF HL Transmissions, Hydraulics), • Replenish oils and fluids (Engine, ZF HL Transmissions, axles, hydraulic system and others), • Check that machine components (dimensions, flatness, voltage, current, tightness, pressure, etc.) comply with the manufacturer's recommendations, • Remove and refit components (filters, belts and accessories, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the systems on tracked excavators JS100W series, • Reset the maintenance counter after operations in accordance with the manufacturer's recommendations. ► **Evaluation part :** • Initial test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	E-Learning and/or classroom training in Mechanics, Electricity and Basic Hydraulics		
Language	French		
Equipment	Fully-equipped training room and workshop, Adapted machine(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

JCB05	PREVENTIVE MAINTENANCE OF TELESCOPIC HANDLERS JCB 540-170 SERIES

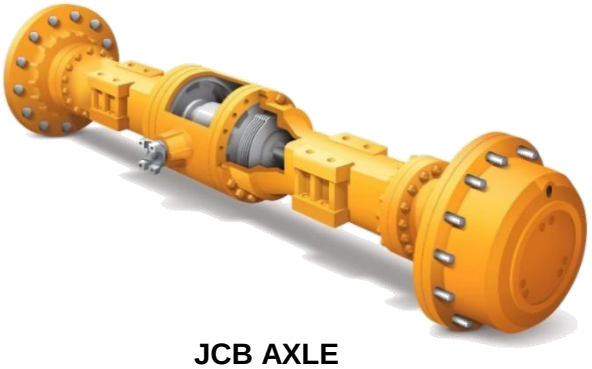
Level, mode & duration	JUNIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Maintain** the systems (powertrain, drive train, chassis, electricity, hydraulics) in operation in accordance with the manufacturer's recommendations. • **Inspect** the machine's systems (powertrain, chassis, electricity, hydraulics) in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Ranges of telescopic handlers JCB 540-170 Series and their characteristics, • Description of machine systems (Powertrain, Chassis, electricity, hydraulics, air conditioning, Steering, ...) and their components, • The different types of maintenance applied to telescopic handlers JCB 540-170 Series and maintenance periods in accordance with the manufacturer's recommendations, • Periodic and non-periodic preventive maintenance of systems and components in accordance with the manufacturer's recommendations. ► **Practical part :** • Replace consumables (Engine, Transmissions, Hydraulics), • Replenish oils and fluids (Engine, Transmissions, axles, hydraulic system and others), • Check that machine components (dimensions, flatness, voltage, current, tightness, pressure, etc.) comply with the manufacturer's recommendations, • Remove and refit components (filters, belts and accessories, hydraulic hoses, etc.) in accordance with the manufacturer's recommendations, • Visually inspect the telescopic handler truck systems JCB 540-170 Series, • Reset the maintenance counter after operations in accordance with the manufacturer's recommendations. ► **Evaluation part :** • Initial test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	Training (e-learning and/or face-to-face) in basic mechanics, electricity and hydraulics		
Language	English - French		
Equipment	Fully-equipped training room and workshop, adapted machine(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		

SENIOR TECHNICIAN

REPAIR



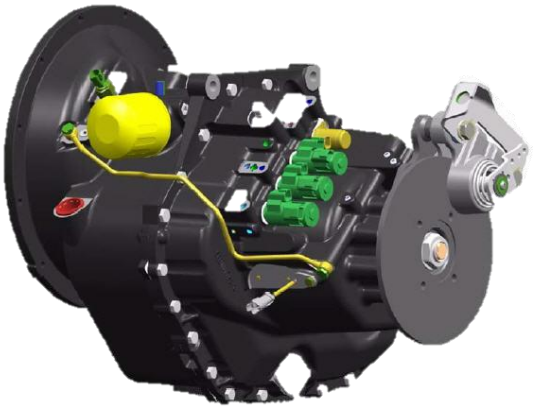
DIESEL MAX ENGINE



JCB AXLE



SYNCHROSHIFT GEARBOX



POWERSHIFT GEARBOX



HYDRAULIC CONTROL



POWERSHIFT WG-Series TRANSMISSION

JCB07	REPAIR OF POWERSHIFT PS750/764/766 GEARBOX

Level, mode & duration	SENIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the POWERSHIFT gearbox in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the gearbox POWERSHIFT in accordance with the manufacturer's recommendations, • **Check** that the components comply with the manufacturer's recommendations, • **Test** and **adjust** the POWERSHIFT gearbox to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Description and operation of POWERSHIFT(PS750/764/766) gearbox, • Description of the electrohydraulic gearshift control circuit, • Procedure for removal and refit of a POWERSHIFT gearbox in accordance with the manufacturer's recommendations, • Procedure for disassembly and reassembly of the components of a POWERSHIFT gearbox, • Procedure for checking the components of a POWERSHIFT gearbox in accordance with the manufacturer's recommendations. ► **Practical part :** • Remove the POWERSHIFT gearbox and the electrohydraulic valve control unit in accordance with the manufacturer's recommendations, • Disassemble the POWERSHIFT gearbox in accordance with the manufacturer's recommendations, • Check that the components (electro-hydraulic valves, clutch blocks, etc.) comply with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the POWERSHIFT gearbox in accordance with the manufacturer's recommendations, • Test and adjust the POWERSHIFT gearbox. ► **Evaluation part :** • Initial Test (Assessment of Pre-requisites) • Final Test (Assessment of Acquired skills)		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Maintenance)		
Language	French - English		
Equipment	Fully-equipped training room and workshop, JCB machine(s) fitted with a POWERSHIFT gearbox, POWERSHIFT Gearbox on bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

JCB08	REPAIR OF SYNCHROSHIFT SS 600/700 GEARBOX

Level, mode & duration	SENIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the SYNCHROSHIFT gearbox in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the SYNCHROSHIFT gearbox in accordance with the manufacturer's recommendations, • **Check** the component conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the SYNCHROSHIFT gearbox to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Description and operation of the SYNCHROSHIFT gearbox (600 and 700 Series), • Description of the electrohydraulic gearshift control circuit, • Procedure for removal and refit of SYNCHROSHIFT gearbox according to the manufacturer's recommendations, • Procedure for disassembly and reassembly of SYNCHROSHIFT gearbox components in accordance with the manufacturer's recommendations, • Procedure for checking SS gearbox components in accordance with the manufacturer's recommendations. ► **Practical part :** • Remove the SYNCHROSHIFT gearbox and Easy Shift Transmission (EST) control unit in accordance with the manufacturer's recommendations, • Disassemble the SYNCHROSHIFT gearbox according to the manufacturer's recommendations, • Check that the components (hydraulic solenoid valves, clutch units, synchronizers, etc.) comply with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the SYNCHROSHIFT gearbox in accordance with the manufacturer's recommendations, • Test and adjust the SYNCHROSHIFT Gearbox, ► **Evaluation part :** • Initial Test (Assessment of Pre-requisites) • Final Test (Assessment of Acquired skills)		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Maintenance)		
Language	English - French		
Equipment	Fully-equipped training room and workshop, JCB machine fitted with SYNCHROSHIFT gearbox, SYNCHROSHIFT gearbox on bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

JCB09	REPAIR OF POWER SHIFT 3/4/6WG ERGOPOWER GEARBOX

Level, mode & duration	SENIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the POWERSHIFT WG ERGOPOWER gearbox in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the POWERSHIFT WG ERGOPOWER(3/4/6WG) gearbox in accordance with the manufacturer's recommendations, • **Check** the component's conformity in accordance with the manufacturer's recommendations, • **Test** and **adjust** the POWERSHIFT WG ERGOPOWER gearbox according to the manufacturer's recommendations.		
Description Théorie (40%) Pratique (60%)	► **Theoretical part :** • Description and operation of POWERSHIFT ERGOPOWER (3/4/6WG) gearbox, • Description of the electrohydraulic gearshift control circuit, • Procedure for removal and refit of a POWERSHIFT ERGOPOWER (3/4/6WG) gearbox according to the manufacturer's recommendations, • Procedure for disassembly and reassembly of POWERSHIFT ERGOPOWER (3/4/6WG) gearbox components in accordance with the manufacturer's recommendations, • Procedure for checking SS gearbox components in accordance with the manufacturer's recommendations. ► **Practical part :** • Remove the POWERSHIFT ERGOPOWER (3/4/6WG) gearbox and ERGOCONTROL command unit in accordance with the manufacturer's recommendations, • Disassemble the POWERSHIFT ERGOPOWER (3/4/6WG) gearbox according to the manufacturer's recommendations, • Check that the components (hydraulic solenoid valves, clutch units, synchronizers, etc.) comply with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the POWERSHIFT ERGOPOWER (3/4/6WG) gearbox in accordance with the manufacturer's recommendations, • Test and adjust the POWERSHIFT ERGOPOWER (3/4/6WG) gearbox, ► **Evaluation part :** • Initial Test (Assessment of Pre-requisites) • Final Test (Assessment of Acquired skills)		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Maintenance)		
Language	French		
Equipment	Fully-equipped training room and workshop, JCB machine fitted with POWERSHIFT ERGOPOWER (3/4/6WG) gearbox, POWERSHIFT ERGOPOWER (3/4/6WG) gearbox on bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

JCB10	REPAIR OF DIESELMAX 444 TIER 2/3 ENGINE WITH MECHANICAL AND ELECTRONIC INJECTION

Level, mode & duration	SENIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the Dieselmax 444 engine in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the Dieselmax 444 engine in accordance with the manufacturer's recommendations, • **Check** that the components comply with the manufacturer's recommendations, • **Test** and **adjust** the Dieselmax 444 engine in accordance with the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Description and operation of the Dieselmax 444 engine, • Description of the lubrication and cooling systems, • Description of the fuel supply system, • Description of the electronic engine management system, • Removal and refit procedure in accordance with the manufacturer's recommendations, • Disassembly and reassembly procedure in accordance with the manufacturer's recommendations, • Procedure for checking engine components in accordance with the manufacturer's recommendations. ► **Practical part:** • Remove the Dieselmax 444 engine according to the manufacturer's recommendations, • Disassemble the Dieselmax 444 engine according to the manufacturer's recommendations, • Clean the Dieselmax 444 engine components, • Visually and geometrically check the components, • Check that the components comply with the manufacturer's recommendations, • Replace components that do not comply with the manufacturer's recommendations, • Reassemble the components in accordance with the manufacturer's recommendations, • Refit the Dieselmax 444 engine in accordance with the manufacturer's recommendations, • Test and adjust the engine on the machine. ► **Evaluation part:** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Maintenance)		
Language	French		
Equipment	Fully-equipped training room and workshop, JCB machine(s) fitted with Dieselmax 444 engine, Dieselmax 444 engine on bench.		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

JCB11	REPAIR OF HYDRAULICS FOR JCB MACHINES

Level, mode & duration	SENIOR	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Remove** and **refit** the machine's hydraulic components in accordance with the manufacturer's recommendations, • **Disassemble** and **reassemble** the machine's hydraulic system components in accordance with the manufacturer's recommendations, • **Check** that the components comply with the manufacturer's recommendations, • **Test** and **adjust** the hydraulic components and hydraulic circuit parameters to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical part:** • Basic architecture of JCB hydraulic systems and interpretation of a hydraulic diagram, • Description and operation of the hydraulic system of a JCB machine and its basic components (Pump, Motor, distributor and Cylinders, ...), • Removal and refit procedure of hydraulic components (Pump, Engine, Distributor and Cylinders, Servo-joystick, etc.) in accordance with the manufacturer's recommendations, • Disassembly and reassembly procedure of hydraulic components (Pump, Motor, Distributor and Cylinders, Servo-joystick, etc.) in accordance with the manufacturer's recommendations, • Procedure for checking hydraulic components or circuit in accordance with the manufacturer's recommendations. ► **Practical part:** • Remove the hydraulic components in accordance with the manufacturer's recommendations, • Disassemble the hydraulic components in accordance with the manufacturer's recommendations, • Clean the hydraulic components (Pumps, Motors, Distributor, jacks, servo-joysticks, ...), • Check that components comply with the manufacturer's recommendations, • Replace components that do not comply with manufacturer's recommendations, • Reassemble components in accordance with the manufacturer's recommendations, • Refit components in accordance with the manufacturer's recommendations, • Test and adjust hydraulic components. ► **Evaluation part:** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills)		
Target audience	Technician carrying out repairs		
Prerequisites	Have acquired the skills of a JUNIOR Technician (Maintenance)		
Language	French		
Equipment	Fully-equipped training room and workshop, adapted JCB machine(s) available, SST		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation.		

EXPERT TECHNICIAN

DIAGNOSTIC



JCB Servicemaster 4 v21.3.3







JCB12	ELECTRICAL/ELECTRONIC DIAGNOSTIC OF JCB MACHINES

Level, mode & duration	Expert	Face-to-face	05 Days
Trainer	**Thierry KOUGHA**		
Objectives	• **Diagnose** faults relating to the electrical and electronic equipment of JCB machines • **Recommend** repair actions.		
Description Theory (40%) Practical (60%)	► **Theoretical part :** • Description and maintenance reminder of the electronic management system of JCB machines, • Electrical/Electronic Architecture of Machines and Interconnection of modules on the CAN network, • Description and operation of the electronic management systems on JCB machines (WECU Transmission Management, HECU Hydraulic Management, MECU Machine Management, DECU Instrument cluster management), • Malfunctions in MECU, HECU, DECU and WECU electronic management systems, • Interconnection of the MECU, WECU, DECU and HECU electronic modules on the machine's CAN network, • Setting machine parameters from the DECU Smart Control instrument cluster, • Diagnostic steps and functions of the SM4 + DLA 2.0 diagnostic tool (Component calibration, Machine parameter setting, Module programming). ► **Practical part:** 1. Collect the various symptoms of the malfunction/fault and the history of breakdowns and repairs on the machine (JCB Service Master 4). 2. Hypothesize the cause of the machine malfunction/fault, 3. Test or check the conformity of the machine's system and components (Service Master 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Elaborate a list of components/organs to be changed or repaired, 6. Repair the faulty component or organ, 7. Test and adjust the system to be repaired (JCB Service Master 4). ► **Evaluation part:** • Initial Test (Assessment of Prerequisites). • Final Test (Assessment of Acquired skills).		
Target audience	Diagnostic technician		
Prerequisites	Have acquired the skills of a SENIOR Technician (Repair)		
Language	English - French		
Equipment	Fully-equipped training room and workshop, suitable machine(s) available, diagnostic interface, special tools		
Documentation	Attendance sheet, Satisfaction sheet, Attestation of participation		